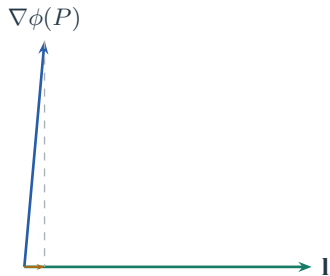
例：方向导数是梯度的有向投影



给定方向几乎垂直于梯度，正投影较小

$$\nabla\phi(P) = (-2,1,3)$$
$$\mathbf{l} = (2,-1,2)/3,\ |\mathbf{l}| = 1$$

$$D_{\mathbf{l}}\phi(P) = \nabla\phi(P)\cdot\mathbf{l} = 1/3$$

$$|\nabla\phi(P)| = \sqrt{14}$$

给定方向的变化率小于最大增长率。